1 sum of two matrices:

code

```
#include<stdio.h>
int main(){
int r,c,i,j;
printf("Enter rows and columns:");
scanf("%d%d",&r,&c);
int a[10][10],b[10][10],sum[10][10];
printf("Enter elements of 1st matrix:\n");
for(i=0;i<r;i++){
for(j=0;j<c;j++){
scanf("%d",&a[i][j]);
}
}
printf("Enter elements of 2nd matrix:\n");
for(i=0;i<r;i++){
for(j=0;j<c;j++){
scanf("%d",&b[i][j]);
}
}
for(i=0;i<r;i++){
for(j=0;j<c;j++){
sum[i][j]=a[i][j]+b[i][j];
}
}
printf("Sum of matrices:\n");
for(i=0;i<r;i++){
for(j=0;j<c;j++){
printf("%d ",sum[i][j]);
}
printf("\n");
return 0;
}
```

output:

```
Enter rows and columns:2# #2
1
Enter elements of 1st matrix:
1
4
Enter elements of 2nd matrix:
6
4
Sum of matrices:
7
8
```

2.Write a program in C to read n number of values in an array & display it in reverse order.

code

```
#include<stdio.h>

int main()
{
    int a[100], n, i;

    printf("Enter number of elements:");
    scanf("%d",&n);
```

```
    printf("Enter %d elements:",n);

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    printf("Array in reverse order:\n");

    for(i=n-1;i>=0;i--)
    {
        printf("%d ",a[i]);
    }

    return 0;
}
```

output:

```
Enter number of elements:4
Enter 4 elements:1 2 3 4
Array in reverse order:
4 3 2 1
```

3.Count duplicate elements in an array

code

```
#include <stdio.h>
int main()
{
    int a[100], n, i, j, count = 0;

    printf("Enter number of elements:");
    scanf("%d",&n);

    printf("Enter %d elements:\n",n);

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    for(i=0;i<n;i++)
    {
        for(j=i+1;j<n;j++)
        {
            if(a[i]==a[j])
            {
                count++;
                break;
            }
        }
    }

    printf("Total duplicate elements:%d\n",count);

    return 0;
}
```

output

```
Enter number of elements:6
```

```
Enter 6 elements:
1 2 3 2 4 1
Total duplicate elements:2
```

4. Print unique elements in an array

code

```
#include <stdio.h>

int main()
{
    int a[100], n, i, j, count;

    printf("Enter number of elements:");
    scanf("%d",&n);

    printf("Enter elements:\n");

    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    printf("Unique elements are:");

    for(i=0;i<n;i++)
    {
        count=0;

        for(j=0;j<n;j++)
        {
            if(a[i]==a[j])
            {
                count++;
            }
        }

        if(count==1)
        {
            printf("%d",a[i]);
        }
    }

    return 0;
}
```

output

```
Enter number of elements:6
Enter elements:
1 2 2 3 4 4
Unique elements are:13
```

5. Insert element in sorted array

code

```
#include<stdio.h>
int main()
{
    int n,i,element;

    printf("Enter number of elements:");
```

```
    scanf("%d",&n);

    int arr[100];

    printf("Enter sorted elements:");
    for(i=0;i<n;i++)
    {
        scanf("%d",&arr[i]);
    }

    printf("Enter element to insert:");
    scanf("%d",&element);

    for(i=n-1;i>=0&&arr[i]>element;i--)
    {
        arr[i+1]=arr[i];
    }

    arr[i+1]=element;
    n++;

    printf("Array after insertion:");
    for(i=0;i<n;i++)
    {
        printf("%d ",arr[i]);
    }

    return 0;
}
```

output

```
Enter number of elements:5
Enter sorted elements:1 3 5 7 9
Enter element to insert:6
Array after insertion:1 3 5 6 7 9
```

6. Insert an element in an unsorted array

code

```
#include<stdio.h>
int main()
{
    int a[100],n,pos,ele,i;

    printf("Enter size:");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }

    printf("Enter position:");
    scanf("%d",&pos);

    printf("Enter element:");
    scanf("%d",&ele);

    for(i=n;i>=pos;i--)
    {
        a[i]=a[i-1];
    }
```

```
    a[pos-1]=ele;
    n++;

    printf("Array after insertion:\n");
    for(i=0;i<n;i++)
    {
        printf("%d ",a[i]);
    }

    return 0;
}
```

output:

```
Enter size:5
Enter elements:
10  # #20 30 40 50
Enter position:3
Enter element:25
Array after insertion:
10 20 25 30 40 50
```

7. Matrix Multiplication

code

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], c[10][10];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of matrix A:");
    scanf("%d%d", &r1, &c1);

    printf("Enter rows and columns of matrix B:");
    scanf("%d%d", &r2, &c2);

    if(c1 != r2)
    {
        printf("Matrix multiplication not possible");
        return 0;
    }

    printf("Enter elements of matrix A:\n");
    for(i=0; i<r1; i++)
    {
        for(j=0; j<c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter elements of matrix B:\n");
    for(i=0; i<r2; i++)
    {
        for(j=0; j<c2; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i=0; i<r1; i++)
```

```
    {
        for(j=0; j<c2; j++)
        {
            c[i][j] = 0;

            for(k=0; k<c1; k++)
            {
                c[i][j] = c[i][j] + a[i][k] * b[k][j];
            }
        }
    }

    printf("Result Matrix:\n");

    for(i=0; i<r1; i++)
    {
        for(j=0; j<c2; j++)
        {
            printf("%d\t", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

output:

```
Enter rows and columns of matrix A:2 2
Enter rows and columns of matrix B:2 2
Enter elements of matrix A:
1 2
3 4
Enter elements of matrix B:
5 6
7 8
Result Matrix:
19    22
43    50
```

8. Matrix Transpose

code

```
#include<stdio.h>
int main()
{
    int a[10][10],t[10][10],r,c,i,j;

    printf("Enter rows and columns:");
    scanf("%d%d",&r,&c);

    printf("Enter elements:");
    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
        {
            scanf("%d",&a[i][j]);
        }
    }

    for(i=0;i<r;i++)
    {
        for(j=0;j<c;j++)
        {
```

```
            t[j][i]=a[i][j];
        }
    }

    printf("Transpose matrix:\n");
    for(i=0;i<c;i++)
    {
        for(j=0;j<r;j++)
        {
            printf("%d ",t[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

output:
```
Enter rows and columns:2 2 2# #
Enter elements:1 2
3 4
Transpose matrix:
1 3
2 4
```

9.find right diagonal sum

code

```
#include <stdio.h>
int main()
{
    int a[10][10], n, i, j, sum = 0;

    printf("Enter order of matrix:");
    scanf("%d", &n);

    printf("Enter matrix:\n");

    for(i = 0; i < n; i++)
        for(j = 0; j < n; j++)
            scanf("%d", &a[i][j]);

    for(i = 0; i < n; i++)
        sum += a[i][n - i - 1];

    printf("Right diagonal sum = %d", sum);

    return 0;
}
```

output:

```
Enter order of matrix:3
Enter matrix:
1 2 3
 4^[[B# ## ## ## ## ## #4 5 6
7 8 9
Right diagonal sum = 15
```

10. check you an identity matrix

code

```c
#include <stdio.h>
int main()
{
    int a[10][10], n, j, i, flag=1;

    printf("Enter order of matrix:");
    scanf("%d", &n);

    printf("Enter matrix:\n");

    for(i=0; i<n; i++)
        for(j=0; j<n; j++)
            scanf("%d", &a[i][j]);

    for(i=0; i<n; i++)
    {
        for(j=0; j<n; j++)
        {
            if(i==j && a[i][j]!=1)
                flag=0;

            if(i!=j && a[i][j]!=0)
                flag=0;
        }
    }

    if(flag)
        printf("Identity matrix");
    else
        printf("Not an identity matrix");

    return 0;
}
```

output:

```
Enter order of matrix:3
Enter matrix:
1 0 0
0 1 0
0 0 1
Identity matrix
```

11. Count vowels, consonants, digits and white spaces

code

```c
#include <stdio.h>
#include <ctype.h>

int main()
{
    char str[200];
    int i, vowels = 0, consonants = 0;
    int digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(isalpha(str[i]))
        {
            if(str[i] == 'a' || str[i] == 'e' ||
```

```
               str[i] == 'i' || str[i] == 'o' ||
               str[i] == 'u' ||
               str[i] == 'A' || str[i] == 'E' ||
               str[i] == 'I' || str[i] == 'O' ||
               str[i] == 'U')
            {
                vowels++;
            }
            else
            {
                consonants++;
            }
        }
        else if(isdigit(str[i]))
        {
            digits++;
        }
        else if(str[i] == ' ')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White spaces = %d\n", spaces);

    return 0;
}
```

output:

```
Enter a string: hello world 123
Vowels = 3
Consonants = 7
Digits = 3
White spaces = 2
```

12. Find frequency of a character in a string

code

```
#include <stdio.h>

int main()
{
    char str[200], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    printf("Enter character: ");
    scanf(" %c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
            count++;
    }

    printf("frequency of '%c' = %d", ch, count);
```

```
    return 0;
}
```

output:

```
Enter a string: banana
Enter character: a
frequency of 'a' = 3
```

13. Count words in a string

code

```
#include <stdio.h>

int main()
{
    char str[200];
    int i, words = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if((i == 0 && str[i] != ' ' && str[i] != '\n') ||
           (str[i - 1] == ' ' && str[i] != ' ' && str[i] != '\n'))
        {
            words++;
        }
    }

    printf("Number of words = %d", words);

    return 0;
}
```

output:

```
Enter a string: c programming is easy
Number of words = 4
```

14. Sort string array

code

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][50], temp[50];
    int n, i, j;

    printf("Enter number of strings: ");
    scanf("%d", &n);

    printf("Enter strings:\n");

    for(i = 0; i < n; i++)
        scanf("%s", str[i]);

    for(i = 0; i < n - 1; i++)
```

```
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Sorted strings:\n");

    for(i = 0; i < n; i++)
        printf("%s\n", str[i]);

    return 0;
}
```

output:

```
Enter number of strings: 3
Enter strings:
mango
apple
banana
Sorted strings:
apple
banana
mango
```

15. Toggle case of a sentence

code

```
#include <stdio.h>
#include <ctype.h>

int main()
{
    char str[200];
    int i;

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(islower(str[i]))
            str[i] = toupper(str[i]);
        else if(isupper(str[i]))
            str[i] = tolower(str[i]);
    }

    printf("After toggling case: %s", str);

    return 0;
}
```

output:

```
Enter a sentence: Hello world
```

```
After toggling case: hELLO WORLD
```